

Competitive screening in labour markets

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MSPC Chapter 4.3, although this lecture departs somewhat from the book, so read through the slides before the lecture.

Competitive screening

- The uninformed parties develop mechanisms to screen, or distinguish, informed individuals.
- The screening we consider here is competitive, since we assume there are several (more than one) firms competing for workers.
- Key to being able to screen: Firms must be able to offer something that is of different attractiveness to different types of workers.
- If the different types of workers have different costs (disutilities) of what the firm offers/requires them to do, then firms can offer a menu of contracts where each contract will be the most attractive for a specific type.

Model

- There are two types of workers: $\theta \in \{\theta_L, \theta_H\}$, with $\theta_H > \theta_L > 0$. The worker's type is his ability.
- The fraction of workers who are of type θ_H is $\lambda \in (0, 1)$ (alternatively assume there is one worker who is of type θ_H with prob. λ).
- There are two firms that do not observe the worker's type.
- Firms can require workers to engage in a level $t \geq 0$ of a (useless) task. The only effect of a higher task level is to lower the utility of the workers.
- A contract is a wage-task pair (w, t) . So jobs can differ in the task level required of the worker.

Timing

Two stage game:

Stage 1: The two firms simultaneously announce any finite number of contracts (which are (w, t) pairs). For example, a firm might offer two contracts: one which pays a high wage, but requires high task level and one which pays a low wage, but requires low task level.

Stage 2: Given the contracts offered by the firms, workers of each type choose whether to accept a contract and if so, which one.

- If indifferent, workers choose the contract with lower t .
- Workers accept employment if indifferent between work and outside option.
- If the worker's most preferred contract is offered by both firms, she randomizes 50-50.

Payoffs

- The profit of a firm if it hires a worker of type θ and pays wage w is

$$\Pi(\theta, w) = \theta - w.$$

Note that we assume that the level of the task does not affect the worker's productivity.

- Worker with productivity θ who accepts contract with wage w and task level t has utility

$$U(\theta, t, w) = w - c(t, \theta)$$

If the worker does not accept any contract he has utility zero.

Assumptions

$$c(0, \theta) = 0 \quad \forall \theta \in \{\theta_L, \theta_H\}$$

$$c_t(t, \theta) > 0 \quad \forall t > 0, \forall \theta \in \{\theta_L, \theta_H\}$$

$$c_{tt}(t, \theta) > 0 \quad \forall \theta \in \{\theta_L, \theta_H\}$$

$$c(t, \theta_L) > c(t, \theta_H) \quad \forall t > 0$$

$$c_t(t, \theta_L) > c_t(t, \theta_H) \quad \forall t > 0$$

where $c_t(t, \theta) = \frac{\partial c(t, \theta)}{\partial t}$.

- The key assumption is “single crossing”: low type has higher costs and higher marginal costs than high type.

- We have a game where the players are the two firms and the worker.
- We look for SPNE.
- We will illustrate key features under the first and second best graphically.
- We will proceed in a number of steps essentially eliminating different types of potential equilibria by arguing that “profitable deviations exists”. This implies firms would have wanted to act differently and hence it cannot be an equilibrium in the first case.
 - finding profitable deviations (modstrid) is a very powerful strategy towards eliminating potential equilibria in this class of models!

First best: complete information

Result: In any SPNE of the screening game with observable types a type $\theta_i \in \{\theta_L, \theta_H\}$ worker accepts contract $(w_i^{FB}, t_i^{FB}) = (\theta_i, 0)$ and firms earn zero profits.

Proof:

- ① Suppose $w_i^{FB} > \theta_i$:

What can one conclude for the profits of at least one of the firms? Can this firm improve?

- ② Suppose $w_i^{FB} < \theta_i$:

Let Π be the aggregate profit. Since $w < \theta$, $\Pi > 0$.

One firm is earning no more than $\frac{\Pi}{2}$.

Can you think of a profitable deviation?

First best: Complete information

The SPNE has $t=0$. Why?

Are there profitable deviations from (θ_i, t') for $t' > 0$?

First best: Complete information

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Intuition:

Under complete information all of the assumptions behind the fundamental welfare theorems are satisfied.

Thus the market outcome is efficient.

The task is useless.

Asymmetric information

- Now each contract offered by a firm can in principle be accepted by either type.
- What happens if firms offer the first best contracts (i.e. the solution from complete information)?
To the high type: $(w_H^{FB}, t_H^{FB}) = (\theta_H, 0)$
To the low type: $(w_L^{FB}, t_L^{FB}) = (\theta_L, 0)$
- Type θ_L prefers type θ_H 's contract.
- What can P do? Under asymmetric info, there are two types of outcomes: Separating or pooling equilibria.

We study two types of contracts (pooling and separating) and look for SPNE. Solving for SPNE will consist of a series of sub-results:

- 1 In any equilibrium (pooling or separating) both firms must earn zero profits.
- 2 No pooling equilibria exist. Therefore any equilibrium must be a separating equilibrium. So we focus on these.
- 3 If (w_L, t_L) and (w_H, t_H) are the contracts signed by the low- and high-ability workers in a separating equilibrium, then $w_L = \theta_L$ and $w_H = \theta_H$, i.e. both contracts yield zero profits.
- 4 In any separating equilibrium the low-ability workers accept contract $(\theta_L, 0)$; i.e. they receive the same contract as when no informational asymmetries are present.
- 5 In any separating equilibrium the high-ability workers accept contract $(\theta_H, \hat{t}_H)$, where $\hat{t}_H$ satisfies $\theta_H - c(\hat{t}_H, \theta_L) = \theta_L - c(0, \theta_L)$.

Result 1

In any equilibrium both firms must earn zero profits.

Essentially same argument as above:

If aggregate $\Pi > 0$ then there exists profitable deviations from either firm. By increasing wages a little they attract all workers and increase individual profits. Thereby a profitable deviation exists and we are not in equilibrium.

Similarly profits cannot be negative in equilibrium.

Implication of Result 1

Corollary 1: In any equilibrium, no firm can have a deviation that allows it to earn strictly positive profits.

Result 2

No pooling equilibria exist.

A profitable deviation always exists!

Find a $(\tilde{w}, \tilde{t})$ which attracts θ_H only.

Result 3

If (w_L, t_L) and (w_H, t_H) are the contracts signed by the low- and high-ability workers in a separating equilibrium, then $w_L = \theta_L$ and $w_H = \theta_H$, i.e. both contracts yield zero profits.

Why?

If $w_L < \theta_L$ then either firm will have an incentive to offer a new wage $\tilde{w}_L$ with $w_L < \tilde{w}_L < \theta_L$. If the new wage attracts θ_H , the firm profits even further.

Therefore, $w_L \geq \theta_L$ in a separating equilibrium. As a consequence, $w_H \leq \theta_H$.

What if $w_H < \theta_H$. Is there a profitable deviation?

Result 4

In any separating equilibrium the low-ability workers accept contract $(\theta_L, 0)$.

I.e. the low type receives the same contract as when no informational asymmetries are present.

Why?

In a potential separating equilibrium with $t'_L > 0$ it is always possible to decrease task requirement and wages and still prevent the high type from pretending to be off low type.

Result 5

In any separating equilibrium the high-ability workers accept contract $(\theta_H, \hat{t}_H)$, where $\hat{t}_H$ satisfies $\theta_H - c(\hat{t}_H, \theta_L) = \theta_L - c(0, \theta_L)$.

Suppose $t_H > \hat{t}_H$. Then either firm has an incentive to deviate by offering contract with a slightly lower wage and task $(\tilde{w}, \tilde{t})$. This contract is only accepted by θ_H and as $\tilde{w} < \theta_H$ it earns strictly positive profits and therefore it is not an equilibrium with $t_H > \hat{t}_H$.

Comment: The low types incentive compatibility constraint is

$$\theta_L - c(0, \theta_L) \geq \theta_H - c(t_H, \theta_L).$$

It says that he weakly prefers the contract designed for him, $(\theta_L, 0)$, over the contract designed for type θ_H : (t_H, θ_H) .

In a separating equilibrium we require the IC-constraint to hold.

In any equilibrium of the screening game, low-ability workers accept contract $(\theta_L, 0)$ and high ability workers $(\theta_H, \hat{t}_H)$, where $\hat{t}_H$ satisfies $\theta_H - c(\hat{t}_H, \theta_L) = \theta_L - c(0, \theta_L)$.

Hence, to prevent low ability workers from pretending to be high ability workers, the principal makes high ability workers perform an “unproductive” task.

In the solution above, no firm can earn strictly positive profits by deviating in a manner that attracts either all high-type or all low-type workers (see construction of the separating equilibrium). **But** under some circumstances a contract that breaks the only candidate for equilibrium exists $\rightarrow$ i.e. no equilibrium exists!

Case 1: A deviation to a contract that attracts all workers to a single pooling contract may pay off (the pooling contract $(\tilde{w}, \tilde{p})$ breaks the separating equilibrium)

Case 2: Even if no pooling contract breaks the separating equilibrium, it is possible that a profitable deviation involving a pair of contracts may do.

- Note that both cases require λ to be sufficiently high.

Later: Screening without competition avoids non-existence problems! (monopolistic screening)

Welfare properties

Pareto inefficiencies: high-ability workers accept contracts that make them engage in a completely unproductive tasks ($t_H > 0$) which give them disutility.

- this is done to distinguish them from their less able counterparts.
- the inefficiency is clear because there are firms who would be willing to hire θ_H workers at contracts which make these workers strictly better off. I.e. not all gains from trade are exhausted!
- The imperfect information prevents this trade from happening, because these parties cannot find each other - θ_L would pretend to be a high type!

Constrained Pareto Optimal

- But if an equilibrium exists (see next slides) it is constrained Pareto-optimal: a central authority who cannot observe workers types cannot achieve a Pareto improvement either.
- An allocation is “constrained Pareto optimal” or “second best optimal” if it cannot be Pareto improved by a social planner who is unable to observe the agents private information.
- An allocation that is Pareto optimal is clearly also constrained Pareto optimal